

Total No. of Questions : 4]

SEAT No. :

PE-530

[Total No. of Pages : 2

[6578]-3

S.E. (Civil) (Insem.)

FLUID MECHANICS

(2019 Pattern) (Semester - III) (201003)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data if required.
- 5) Use of logarithmic tables, slide rule, Mollier charts, electronic pocket calculator (nonprogrammable) and steam tables is allowed.

- Q1)** a) Differentiate between: i) Absolute and Gauge Pressure ii) Simple manometer and differential manometer. [5]
- b) State and explain Newton's law of viscosity. [5]
- c) Define gauge pressure, vacuum pressure and absolute pressure. [5]

OR

- Q2)** a) Define Specific weight, dynamic viscosity and capillarity. [5]
- b) Derive an expression for a pressure inside a liquid jet of radius R and surface tension. [5]
- c) What is meant by stability of floating and submerged body and what are the different stability conditions for floating and submerged body. [5]

- Q3)** a) Define the following and give one practical example for each : [5]
- i) Laminar flow
 - ii) Turbulent Flow
 - iii) Steady flow
 - iv) Uniform Flow
 - v) Varied Flow

P.T.O.

- b) Write a short note on :- [5]
- i) Stream line
 - ii) Streak Line
 - iii) Path line
 - iv) Hydrokinetics
 - v) Hydro kinematics
- c) The diameter of a pipe at the section 1 and 2 are 15 cm and 20 cm respectively. Find the discharge through the pipe if velocity of water at section 1 is 4 m/s. Determine also the velocity at section 2. [5]

OR

- Q4)** a) Explain how Bernoulli's theorem, applied to two points in flow, is modified to account for [5]
- i) Loss of head,
 - ii) Installation of pump,
 - iii) Non-uniform velocity variation in pipe.
- b) Explain in brief:- [5]
- i) Velocity Potential
 - ii) Stream Function
- c) Determine the total pressure on a circular plate of diameter 1.5 m, which is placed vertically in water in such a way that centre of plate is 2m below the free surface of water. Find the position of centre of pressure also. [5]

